

# \* Classical Harmonic Oscillator →

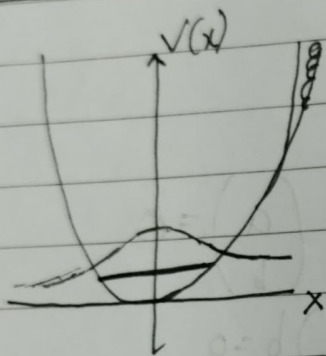
$$F = m \frac{d^2 x}{dt^2} = -kx$$

$$\Rightarrow \frac{d^2 x}{dt^2} + \omega^2 x = 0 \quad ; \quad \omega = \sqrt{\frac{k}{m}}$$

$$x = A \cos \omega t + B \sin \omega t$$

$$V(x) = \frac{1}{2} k x^2 = \frac{1}{2} m \omega^2 x^2$$

$$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 \leftarrow \text{Classical}$$



$$V(x) = \infty \text{ as } x \rightarrow \infty$$

$V(x)$  is even  $\forall x^n$ .

Eigen  $\psi x^n$  are odd or even

G.S. must be even because there can't be any nodes.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \quad ; \quad [\hat{x}, \hat{p}] = i\hbar$$

Schr. Eqn. 
$$-\frac{\hbar^2}{2m} \frac{d^2 \phi}{dx^2} + \frac{1}{2} m \omega^2 x^2 \phi = E \phi$$

$$\Rightarrow \left( \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \right) \phi = E \phi$$

$$\Rightarrow \left( -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 \hat{x}^2 \right) \phi = E \phi$$

Dimensionless variable  $x = \frac{\hbar}{a\sqrt{m\omega}} \xi$  Dimensionless

$$-\frac{\hbar^2}{2m} \frac{d^2 \phi}{dx^2} + \frac{1}{2} m \omega^2 x^2 \phi = E \phi$$

$$\frac{-\hbar^2}{2ma^2}$$

$$\frac{\hbar^2}{2ma^2} = m \omega^2 a^2$$

$$a^2 = \frac{\hbar}{\sqrt{m\omega}}$$

$$a^2 = \frac{\hbar}{m\omega}$$

$$x = \frac{\hbar}{\sqrt{m\omega}} \xi$$

$$-\frac{\hbar^2}{2m} \frac{1}{(\hbar/m\omega)} \frac{d^2 \phi}{d\xi^2} + \frac{1}{2} m \omega^2 \frac{\hbar}{m\omega} \xi^2 \phi = E \phi$$

$$\Rightarrow -\frac{\hbar\omega}{2} \frac{d^2 \phi}{d\xi^2} + \frac{1}{2} \hbar\omega \xi^2 \phi = E \phi$$

$$\Rightarrow -\frac{d^2 \phi}{d\xi^2} + \xi^2 \phi = \left( \frac{2E}{\hbar\omega} \right) \phi \quad E = \frac{2E}{\hbar\omega}$$

Dimensionless

$$\boxed{\frac{d^2 \phi}{d\xi^2} = (\xi^2 - \epsilon) \phi}$$

$$x = \frac{\hbar}{\sqrt{m\omega}} \xi, \quad \epsilon = \frac{2E}{\hbar\omega}$$

For large  $\xi$  ( $\xi \rightarrow \infty$ ); try  $\phi(\xi) = A e^{-\xi^2/2}$

$$\phi(\xi) = h(\xi) e^{-\xi^2/2}$$

$$\frac{d^2 h}{d\xi^2} + \xi^2 h e^{-\xi^2/2} - 2\xi \frac{dh}{d\xi} e^{-\xi^2/2} + (\epsilon - 1) h e^{-\xi^2/2} = 0$$

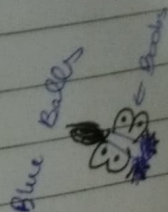
Assume  $h(\xi) = \sum_{k=0}^{\infty} a_k \xi^k \rightarrow$  We need recursion relation

Collect coeff. of  $\xi^j$

$$\frac{d^2 h}{d\xi^2} = (j+2)(j+1) a_{j+2} \xi^j$$

$$-2\xi \frac{dh}{d\xi} = -2j a_j \xi^j$$

$$(\epsilon - 1) h = (\epsilon - 1) a_j \xi^j$$



$$\sum_{j=0}^{\infty} \underbrace{[a_{j+2}(j+2)(j+1) - 2ja_j + (\frac{\epsilon}{2} - 1)a_j]}_0 \xi^j = 0$$

$$(j+2)(j+1)a_{j+2} = (2j+1 - \frac{\epsilon}{2})a_j$$

$$\boxed{a_{j+2} = \frac{(2j+1 - \frac{\epsilon}{2})}{(j+2)(j+1)} a_j}$$

For large  $j$ :  $\frac{a_{j+2}}{a_j} = \frac{2}{j}$

$$e^{\xi^2} = \sum_{j=0}^{\infty} \frac{(\xi^2)^j}{j!} = \sum_{j=0}^{\infty} \frac{\xi^{2j}}{j!} = \sum_{j'=0}^{\infty} \frac{\xi^{j'}}{(j'/2)!}$$

$$C_j = \frac{1}{(j/2)!}$$

$$\frac{C_{j+2}}{C_j} = \frac{(j/2)!}{(j+2/2)!} = \frac{2}{j+2} \rightarrow \frac{2}{j}$$

$$\phi = h e^{-\xi^2/2}$$

$$\hookrightarrow \sim e^{\xi^2} > e^{-\xi^2/2}$$

$\therefore \phi$  won't converge so we must truncate  $h$  (polynomial of degree  $m$ ).

# \* Algebraic Method -

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$$

$$= C \left( \hat{x} + \frac{i\hat{p}}{m\omega} \right) \left( \hat{x} - \frac{i\hat{p}}{m\omega} \right) + B$$

$$\hat{H} = \left( \hat{a}^+ \hat{a} + \frac{1}{2} \right) \hbar \omega$$

$$\hat{N} = \hat{a}^+ \hat{a} \rightarrow \text{Number Operator}$$

$$\hat{H} = \left( \hat{N} + \frac{1}{2} \right) \hbar \omega$$

$$[\hat{N}, \hat{H}] = 0 \rightarrow \hat{N}, \hat{H} \text{ will have a complete set of simultaneous eigenstates.}$$

$$\hat{N} |m\rangle = m |m\rangle$$

$$[\hat{N}, \hat{a}] = [\hat{a}^+ \hat{a}, \hat{a}] = \hat{a}^+ [\hat{a}, \hat{a}] + \underbrace{[\hat{a}^+, \hat{a}]}_{-1} \hat{a}$$

$$= -\hat{a}$$

$$\boxed{[\hat{N}, \hat{a}^+] = \hat{a}^+}$$

$$\boxed{[\hat{N}, \hat{a}] = -\hat{a}}$$

$$\hat{N}(\hat{a}^+ |m\rangle) = (\hat{N} \hat{a}^+ - \hat{a}^+ \hat{N} + \hat{a}^+ \hat{N}) |m\rangle$$

$$= ([\hat{N}, \hat{a}^+] + \hat{a}^+ \hat{N}) |m\rangle$$

$$= (\hat{a}^+ + \hat{a}^+ \hat{N}) |m\rangle$$

$$= \hat{a}^+ |m\rangle + \hat{a}^+ m |m\rangle$$

$$= (m+1) (\hat{a}^+ |m\rangle)$$

Eigen state of  $\hat{N}$   
Eigen value =  $(m+1)$

$$\hat{N} |m+1\rangle = (m+1) |m+1\rangle$$

$$\hat{a}^+ |m\rangle = C_m |m+1\rangle$$

↳ Adds a quantum number

$$\begin{aligned}
 \hat{N} \hat{a} |n\rangle &= ([\hat{N}, \hat{a}] + \hat{a} \hat{N}) |n\rangle \\
 &= (-\hat{a} + \hat{a} \hat{N}) |n\rangle \\
 &= -\hat{a} |n\rangle + \hat{a} n |n\rangle \\
 \hat{N} \hat{a} |n\rangle &= (n-1) (\hat{a} |n\rangle)
 \end{aligned}$$

$$\begin{aligned}
 \hat{N} \hat{a} |n-1\rangle &= a(n-1) |n-1\rangle \\
 \hat{a} |n\rangle &= C_n |n-1\rangle
 \end{aligned}$$

$\hat{a}^+ \rightarrow$  creation operator

$\hat{a} \rightarrow$  annihilation operator

$$\hat{a} |n\rangle = C_n |n-1\rangle$$

$$\langle n | \hat{a}^+ = \langle n-1 | C_n^*$$

$$\begin{aligned}
 (\langle n | \hat{a}^+) (\hat{a} |n\rangle) &= |C_n|^2 \underbrace{\langle n-1 | n-1 \rangle}_{\textcircled{1}} \\
 &= |C_n|^2
 \end{aligned}$$

$$\langle n | \hat{a}^+ \hat{a} |n\rangle = |C_n|^2$$

$$\langle n | \hat{N} |n\rangle = |C_n|^2$$

$$n \langle n | n \rangle \textcircled{1} = |C_n|^2$$

$$\therefore \boxed{C_n = \sqrt{n} e^{+i\phi}}$$

Choosing  $\phi = 0 \Rightarrow$

$$\begin{aligned}
 \hat{a} |n\rangle &= \sqrt{n} |n-1\rangle \\
 \hat{a}^+ |n\rangle &= \sqrt{n+1} |n+1\rangle
 \end{aligned}$$

$$\begin{aligned}
 \langle \Psi | \hat{H} | \Psi \rangle &= \hbar \omega \langle \Psi | \hat{a}^+ \hat{a} | \Psi \rangle + \frac{1}{2} \hbar \omega \\
 &\quad \downarrow (\hat{a}^+ \hat{a} + \frac{1}{2}) \hbar \omega
 \end{aligned}$$

$$\begin{aligned}
 &= (\langle \Psi | \hat{a}^+) (\hat{a} | \Psi \rangle) + \frac{1}{2} \hbar \omega \\
 &= \underbrace{\| \hat{a} | \Psi \rangle \|^2}_{\geq 0} + \frac{1}{2} \hbar \omega
 \end{aligned}$$

Min. value of  $\| \hat{a} | \Psi \rangle \|^2 = 0 \Rightarrow$  If the corresponding state is  $|0\rangle$  then  $\hat{a} |0\rangle = 0$   
 $\hookrightarrow$  Ground State.

$$\hat{a}^+ |m\rangle \sim c_m |m+1\rangle$$

$$\langle m | \hat{a} \sim \langle m+1 | c_m^*$$

$$\langle m | \hat{a} \hat{a}^+ |m\rangle = |c_m|^2$$

$$\hat{a} |m\rangle = \sqrt{m} |m-1\rangle$$

$$\Rightarrow \hat{a}^2 |m\rangle = \hat{a} \sqrt{m} |m-1\rangle$$

$$= \sqrt{m(m-1)} |m-2\rangle$$

$$\Rightarrow \hat{a}^3 |m\rangle = \sqrt{m(m-1)(m-2)} |m-3\rangle$$

The sequence will terminate if we start with +ve integer.  $n$  cannot be non-integer: won't reduce to 0 at  $n=0$

$$|1\rangle = \hat{a}^+ |0\rangle$$

$$|2\rangle = \frac{\hat{a}^+}{\sqrt{2}} |1\rangle \quad \hat{a}^+ |m\rangle = \sqrt{m+1} |m+1\rangle$$

$$|3\rangle = \frac{\hat{a}^+}{\sqrt{3}} |2\rangle = \frac{(\hat{a}^+)^3}{\sqrt{3!}} |0\rangle$$

$$|m\rangle = \frac{(\hat{a}^+)^m}{\sqrt{m!}} |0\rangle$$

$$\hat{H} |m\rangle = \hbar\omega \left( \underbrace{\hat{a}^+ \hat{a}}_{\hat{N}} + \frac{1}{2} \right) |m\rangle = \hbar\omega \left( \hat{N} + \frac{1}{2} \right) |m\rangle$$

$$= \left( m + \frac{1}{2} \right) \hbar\omega |m\rangle$$

$n = 0, 1, 2, \dots$

$$E_m = \left( m + \frac{1}{2} \right) \hbar\omega$$

\* Eigen states in position space →

$a_0(x)$  = ground state in pos<sup>n</sup>. space.

$$\hat{a} \phi_0(x) = 0$$

$$\left( \hat{x} + i \frac{\hat{p}}{m\omega} \right) \phi_0(x) = 0$$

$$x \phi_0(x) + \frac{i}{m\omega} \frac{\hbar}{i} \frac{\partial \phi_0(x)}{\partial x} = 0$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left( \hat{x} + \frac{i\hat{p}}{m\omega} \right)$$

$$x \phi_0(x) + \frac{\hbar}{m\omega} \frac{d\phi_0(x)}{dx} = 0$$

$$\frac{d\phi_0(x)}{dx} = -\frac{m\omega}{\hbar} x \phi_0(x)$$

$$\phi_0(x) = A e^{-\frac{m\omega}{\hbar} x^2}$$

$$\int |\phi_0(x)|^2 dx = 1 \Rightarrow A = \left( \frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}}$$

$$\boxed{\phi_0(x) = \left( \frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{\hbar} x^2}}$$

1st excited state:  $|1\rangle = \frac{(\hat{a}^\dagger)^1}{\sqrt{1!}} |0\rangle$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left( \hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

$$|1\rangle = \hat{a}^\dagger |0\rangle$$

$$= \left( \hat{x} - \frac{i\hat{p}}{m\omega} \right) |0\rangle \sqrt{\frac{m\omega}{2\hbar}}$$

$$= \sqrt{\frac{m\omega}{2\hbar}} \left( \hat{x} - \frac{\hbar}{m\omega} \frac{\partial}{\partial x} \right) |0\rangle$$

$$= \left( \frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} \sqrt{\frac{m\omega}{2\hbar}} \left[ x \phi_2(x) - \frac{\hbar}{m\omega} \frac{d\phi}{dx} \right]$$

$$= \left( \frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} \sqrt{\frac{m\omega}{2\hbar}} \left[ x e^{-\frac{m\omega x^2}{\hbar}} - \frac{\hbar}{m\omega} \left( -\frac{m\omega}{\hbar} \right) \cdot 2x e^{-\frac{m\omega x^2}{\hbar}} \right]$$

$$\phi_1(x) = 2 \left( \frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} \sqrt{\frac{m\omega}{2\hbar}} \underbrace{x e^{-\frac{m\omega x^2}{\hbar}}}_{\text{odd } f(x^m)}$$

$$\phi_m(x) = A_m e^{-\frac{m\omega x^2}{\hbar}} H_m \left( \sqrt{\frac{m\omega}{\hbar}} x \right)$$

Hermite Polynomials.

orthonormality:  $\int_{-\infty}^{\infty} dy e^{-y^2} H_m(y) H_n(y) = c \delta_{mn}$

$$\therefore \int_{-\infty}^{\infty} \phi_m^*(x) \phi_n(x) = 0 \quad ; \quad m \neq n$$

Ans - 6

89.  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \alpha, \beta \rightarrow \text{complex.}$

$$\begin{aligned} \langle \hat{x} \rangle &= \langle \psi | \hat{x} | \psi \rangle \\ &= \langle \alpha^* \langle 0 | + \beta^* \langle 1 | \rangle \hat{x} (\alpha |0\rangle + \beta |1\rangle) \\ &= (\alpha^* \alpha \langle 0 | \hat{x} | 0 \rangle + \alpha^* \beta \langle 0 | \hat{x} | 1 \rangle \\ &\quad + \beta^* \alpha \langle 1 | \hat{x} | 0 \rangle + \beta^* \beta \langle 1 | \hat{x} | 1 \rangle) \end{aligned}$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^*) \quad \begin{aligned} \hat{a}|m\rangle &= \sqrt{m} |m-1\rangle \\ \hat{a}^+|m\rangle &= \sqrt{m+1} |m+1\rangle \end{aligned}$$

$$\begin{aligned} \langle 0 | \hat{x} | 0 \rangle &= \sqrt{\frac{\hbar}{2m\omega}} \langle 0 | \hat{a} + \hat{a}^+ | 0 \rangle \\ &= \sqrt{\frac{\hbar}{2m\omega}} (\langle 0 | \hat{a} | 0 \rangle + \langle 0 | \hat{a}^+ | 0 \rangle) \\ &\quad \langle 0 | 1 \rangle = 0 \end{aligned}$$

$$= 0$$

$$\langle 1 | \hat{x} | 1 \rangle = 0$$

$$\langle \hat{x} \rangle = \alpha^* \beta \langle 0 | \hat{x} | 1 \rangle + \alpha \beta^* \langle 1 | \hat{x} | 0 \rangle$$

$$\langle 0 | \hat{x} | 1 \rangle = \frac{\sqrt{\hbar}}{\sqrt{2m\omega}} \langle 0 | (\hat{a} + \hat{a}^\dagger) | 1 \rangle$$

$$= \frac{\sqrt{2m\omega}}{\sqrt{\hbar}} (\langle 0 | 0 \rangle + \langle 0 | 2 \rangle)$$

$$= \frac{\sqrt{2m\omega}}{\sqrt{\hbar}}$$

$$\langle 1 | \hat{x} | 0 \rangle = \frac{\sqrt{\hbar}}{\sqrt{2m\omega}}$$

$$\langle \hat{x} \rangle_\psi = (\alpha^* \beta + \beta^* \alpha) \frac{\sqrt{\hbar}}{\sqrt{2m\omega}}$$

$$\alpha, \beta \rightarrow \text{real}; |\psi\rangle \text{ normalized} \rightarrow \alpha^2 + \beta^2 = 1$$

$$\langle \hat{x} \rangle_\psi = \frac{\sqrt{\hbar} \sqrt{2} [\alpha \sqrt{1-\alpha^2}]}{\sqrt{2m\omega}}$$

$$= \frac{\sqrt{2\hbar}}{\sqrt{m\omega}} \alpha \sqrt{1-\alpha^2}$$

$$\frac{d\langle \hat{x} \rangle}{d\alpha} = \frac{\sqrt{1-\alpha^2} + \alpha}{\sqrt{1-\alpha^2}} \cdot -2\alpha = 0$$

$$1 - \alpha^2 - \alpha^2 = 0$$

$$\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|\psi(t)\rangle = e^{-\frac{i\hat{H}t}{\hbar}} |\psi(0)\rangle$$

$$= \frac{1}{\sqrt{2}} \left[ e^{-\frac{i\hbar\omega t}{2\hbar}} |0\rangle + e^{-\frac{i3\hbar\omega t}{2\hbar}} |1\rangle \right]$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left[ e^{-\frac{i\omega t}{2}} |0\rangle + e^{-\frac{i3\omega t}{2}} |1\rangle \right]$$

$$\begin{aligned}
 \langle \hat{x} \rangle_{\psi(t)} &= \langle \psi(t) | \hat{x} | \psi(t) \rangle \\
 &= \frac{1}{2} \frac{1}{\sqrt{2}} \left( e^{-\frac{i\omega t}{2}} \langle 0 | + e^{\frac{i\omega t}{2}} \langle 1 | \right) \hat{x} \left( e^{-\frac{i\omega t}{2}} | 0 \rangle + e^{\frac{i\omega t}{2}} | 1 \rangle \right) \\
 &= \frac{1}{2} \langle 0 | \hat{x} | 0 \rangle + e^{-i\omega t} \langle 0 | \hat{x} | 1 \rangle + e^{i\omega t} \langle 1 | \hat{x} | 0 \rangle \\
 &= \frac{1}{2} \sqrt{\frac{\hbar}{2m\omega}} (e^{-i\omega t} + e^{i\omega t}) \\
 &= \sqrt{\frac{\hbar}{2m\omega}} \cos(\omega t)
 \end{aligned}$$

Q-8)  $\langle n | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} \langle n | (\hat{a} + \hat{a}^\dagger)(\hat{a} + \hat{a}^\dagger) | n \rangle$

$$\begin{aligned}
 &= \frac{\hbar}{2m\omega} \left[ \langle n | \hat{a}^2 | n \rangle + \langle n | \hat{a} \hat{a}^\dagger | n \rangle + \langle n | \hat{a}^\dagger \hat{a} | n \rangle + \langle n | \hat{a}^{\dagger 2} | n \rangle \right] \\
 &= \frac{\hbar}{2m\omega} \left[ \sqrt{n+1} \sqrt{n+1} \langle n | n \rangle + \sqrt{n} \sqrt{n} \langle n | n \rangle \right] \\
 &= \frac{(2n+1)\hbar}{2m\omega}
 \end{aligned}$$

$$\begin{aligned}
 \Delta x^2 &= \langle x^2 \rangle - \langle x \rangle^2 \\
 &= \frac{(2n+1)\hbar}{2m\omega} - 0 = \frac{(2n+1)\hbar}{2m\omega}
 \end{aligned}$$

$$\begin{aligned}
 \langle n | \hat{p}^2 | n \rangle &= -\frac{m\omega\hbar}{2} \langle n | (\hat{a}^\dagger - \hat{a})(\hat{a}^\dagger - \hat{a}) | n \rangle \\
 &= \frac{(2n+1)m\omega\hbar}{2}
 \end{aligned}$$

$$\Delta p^2 = \frac{(2n+1)m\omega\hbar}{2}$$

$$\Delta x \Delta p = \left( n + \frac{1}{2} \right) \hbar$$

$$n=0 \Rightarrow \Delta x \Delta p = \frac{\hbar}{2} \Rightarrow \text{Min}^m \text{ uncertainty}$$

2-  $\Psi(x, 0) = \frac{1}{\sqrt{2}} [\Psi_0(x) + i \Psi_1(x)]$

a)  $\Psi(x, t) = \frac{1}{\sqrt{2}} \left[ e^{-\frac{iE_0 t}{\hbar}} \Psi_0(x) + i e^{-\frac{iE_1 t}{\hbar}} \Psi_1(x) \right]$   
 $= \frac{1}{\sqrt{2}} \left[ e^{-\frac{i\omega_0 t}{2}} \Psi_0(x) + i e^{-\frac{i3\omega_0 t}{2}} \Psi_1(x) \right]$

$|\Psi(x, t)|^2 = \Psi^*(x, t) \Psi(x, t)$   
 $= \frac{1}{2} \left[ e^{\frac{i\omega_0 t}{2}} \Psi_0^*(x) - i e^{\frac{i3\omega_0 t}{2}} \Psi_1^*(x) \right] \left[ e^{-\frac{i\omega_0 t}{2}} \Psi_0(x) + i e^{-\frac{i3\omega_0 t}{2}} \Psi_1(x) \right]$   
 $= \frac{1}{2} \left( |\Psi_0(x)|^2 + |\Psi_1(x)|^2 - i e^{i\omega_0 t} \Psi_1^* \Psi_0 + i e^{-i\omega_0 t} \Psi_0^* \Psi_1 \right)$   
 $\Psi_0, \Psi_1 \rightarrow \text{real}$

b)  $\langle \hat{x} \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx$   
 $= \int_{-\infty}^{\infty} x |\Psi_0(x)|^2 dx + \int_{-\infty}^{\infty} x |\Psi_1(x)|^2 dx + 2 \sin \omega_0 t \int_{-\infty}^{\infty} x \Psi_0 \Psi_1 dx$   
 $\begin{matrix} \text{even} & \text{odd} & \text{even} \\ \text{odd} & \text{odd} & \text{odd} \end{matrix}$

$\langle \hat{x} \rangle = \left( \frac{m\omega_0}{\pi \hbar} \right)^{\frac{1}{2}} \frac{2}{\sqrt{2\hbar}} \sin \omega_0 t \int_{-\infty}^{\infty} x^2 e^{-\frac{m\omega_0 x^2}{\hbar}} dx$

$\langle \hat{x} \rangle = \frac{\hbar}{\sqrt{2m\omega_0}} \sin \omega_0 t$

$\langle \hat{p} \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{p} \Psi dx = \int_{-\infty}^{\infty} \Psi^* \frac{\hbar}{i} \frac{\partial}{\partial x} \Psi dx$   
 $= \frac{\hbar}{2i} \left[ \int_{-\infty}^{\infty} \left( e^{\frac{i\omega_0 t}{2}} \phi_0^*(x) - i e^{\frac{3i\omega_0 t}{2}} \phi_1^*(x) \right) \left( e^{-\frac{i\omega_0 t}{2}} \phi_0'(x) + i e^{-\frac{3i\omega_0 t}{2}} \phi_1'(x) \right) dx \right]$

$\phi_0^* = \phi_0$   
 $\phi_1^* = \phi_1$

$\langle \hat{p} \rangle = \frac{\hbar}{2} \int \phi_0 \phi_0' = 0 + \int \phi_1 \phi_1' = 0$

$$\langle \hat{p} \rangle = \frac{\hbar}{2i} \int_{-\infty}^{\infty} [i e^{i\omega_0 t} \phi_1 \phi'_0 + i e^{-i\omega_0 t} \phi_0 \phi'_1] dx$$

$$= \sqrt{\frac{m \hbar \omega_0}{2}} \cos \omega_0 t$$

$$\psi(x, t) = \sum c_n \phi_n(x) e^{-\frac{i E_n t}{\hbar}}$$

$$= e^{-\frac{i \omega_0 t}{2}} \sum c_n \phi_n e^{-i m \omega_0 t}$$

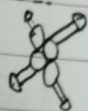
$$E_n = \left( n + \frac{1}{2} \right) \hbar \omega$$

Each component oscillates with time period  $T = \frac{2\pi}{\omega}$

$$T_0 = \frac{2\pi}{\omega_0}, T_1 = \frac{2\pi}{2\omega_0}, T_2 = \frac{2\pi}{3\omega_0}$$

$$\psi(x, t+T) = e^{-\frac{i \omega_0 T}{2}} \psi(x, t)$$

$$|\psi(x, t+T)|^2 = |\psi(x, t)|^2$$



### \* Variational Method →

- ① Pick a trial wave -  $f(x)$ .
- ② Calculate expectation value of Energy in state  $\psi$
- ③  $\langle \hat{H} \rangle_\psi \geq E_g$   $\frac{d\langle \hat{H} \rangle}{db} = 0$

$$\hat{H} \psi_m = E_m \psi_m$$

$$\text{Any state } \psi = \sum c_n \psi_n$$

$$\langle \psi | \psi \rangle = \left\langle \sum_m c_m \psi_m \middle| \sum_n c_n \psi_n \right\rangle$$

$$= \sum_m \sum_n c_m^* c_n \underbrace{\langle \psi_m | \psi_n \rangle}_{\delta_{mn}}$$

$$= \sum |c_n|^2$$

$$\langle \hat{H} \rangle_\psi = \left\langle \sum_m c_m \psi_m \middle| \hat{H} \sum_n c_n \psi_n \right\rangle$$

$$= \left\langle \sum_m c_m \psi_m \middle| \sum_n E_n c_n \psi_n \right\rangle$$

$$= \sum_m \sum_n c_m^* c_n E_n \langle \psi_m | \psi_n \rangle$$

$$= \sum_n |c_n|^2 E_n$$

$$\sum |C_n|^2 E_n \geq E_{gs} \sum |C_n|^2$$

$$E_{gs} \leq E_n$$

$$\langle \Psi | H | \Psi \rangle = \sum |C_n|^2 E_n \geq E_{gs} \underbrace{\sum |C_n|^2}_{(1)}$$

$$\geq E_{gs}$$

$$E_{gs} \leq \langle \hat{H} \rangle_\Psi$$

⇒ 1-D Harmonic Oscillator :-

$$\hat{H} = \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

Use a trial wave  $f(x) \rightarrow \Psi(x) = A e^{-bx^2}$

Normalize  $\Psi(x)$

$$\langle \Psi | \Psi \rangle = |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} dx$$

$$\int_{-\infty}^{\infty} e^{-x^2/a^2} dx = \sqrt{\pi} \frac{(2m)!}{m!} \left(\frac{a}{2}\right)^{2m+1}$$

$$m=0; \quad a = \frac{1}{\sqrt{2}b}$$

$$\langle \Psi | \Psi \rangle = \frac{1}{2} |A|^2 \sqrt{\pi} \frac{1}{\sqrt{2}b} \times \frac{1}{2} = \left(\frac{\pi}{2b}\right)^{1/2} |A|^2$$

$$A = \left(\frac{2b}{\pi}\right)^{1/4}$$

$$\langle \Psi | \hat{H} | \Psi \rangle$$

$$\langle H \rangle = \langle T \rangle + \langle V \rangle$$

$$\langle T \rangle = \frac{-\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} e^{-bx^2} \frac{d^2}{dx^2} (e^{-bx^2}) dx$$

$$= \frac{-\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} e^{-bx^2} \left(\frac{d}{dx}\right)^2 (e^{-bx^2}) dx$$

$$= \frac{\cancel{b} \hbar^2}{2m} \left( \frac{2b}{\pi} \right)^{\frac{1}{2}} \int_{-\infty}^{\infty} e^{-bx^2} \left[ e^{-bx^2} (-2bx^2) + e^{-bx^2} \right] dx$$

$$= \frac{b \hbar^2}{m} \left( \frac{2b}{\pi} \right)^{\frac{1}{2}} \int_{-\infty}^{\infty} (-2bx^2 e^{-2bx^2}) dx + \int_{-\infty}^{\infty} e^{-2bx^2} dx$$

$$\langle T \rangle = \frac{\hbar^2 b}{2m}$$

$$\langle V \rangle = \frac{1}{2} m \omega^2 |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} x^2 dx$$

$$= \frac{m \omega^2}{8b}$$

$$\langle H \rangle = \langle T \rangle + \langle V \rangle = \frac{\hbar^2 b}{2m} + \frac{m \omega^2}{8b} \geq E_{gs}$$

$$\frac{d\langle H \rangle}{db} = \frac{\hbar^2}{2m} - \frac{m \omega^2}{8b^2} = 0$$

$$\frac{\hbar^2}{2m} = \frac{m \omega^2}{8b^2}$$

$$b^2 = \frac{m^2 \omega^2}{4 \hbar^2}$$

$$b = \frac{m \omega}{2 \hbar}$$

$$\langle H \rangle = \frac{\hbar^2}{2m} \cdot \frac{m \omega}{2 \hbar} + \frac{m \omega^2}{8} \cdot \frac{2 \hbar}{m \omega}$$

$$= \frac{\hbar \omega}{2}$$

⇒ Delta Fx<sup>n</sup>. Potential

$$\hat{H} = \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} - \alpha \delta(x)$$

$$\psi(x) = \left(\frac{2b}{\pi}\right)^{\frac{1}{4}} e^{-bx^2}$$

$$\langle T \rangle = \frac{\hbar^2 b}{2m}$$

$$\langle V \rangle = -\alpha \left(\frac{2b}{\pi}\right)^{\frac{1}{2}} \underbrace{\int_{-\infty}^{\infty} e^{-2bx^2} \delta(x) dx}_{(1)}$$

$$= -\alpha \sqrt{\frac{2b}{\pi}}$$

$$\langle A \rangle = \langle T \rangle + \langle V \rangle$$

$$= \frac{\hbar^2 b}{2m} - \alpha \sqrt{\frac{2b}{\pi}}$$

$$\frac{d\langle \hat{H} \rangle}{db} = \frac{\hbar^2}{2m} - \alpha \frac{\sqrt{2} \times \frac{1}{\sqrt{\pi}}}{\sqrt{2} \sqrt{b}} = 0$$

$$\frac{\alpha}{\sqrt{2\pi b}} = \frac{\hbar^2}{2m}$$

$$\frac{2m\alpha}{\hbar^2} = \sqrt{2\pi b}$$

$$\frac{4m^2 \alpha^2}{\hbar^4 \times 2\pi} = b$$

$$b = \frac{2m^2 \alpha^2}{\pi \hbar^4}$$

$$\langle A \rangle = \frac{\hbar^2}{2m} \cdot \frac{2m^2 \alpha^2}{\pi \hbar^4} - \alpha \sqrt{\frac{2}{\pi} \cdot \frac{2m^2 \alpha^2}{\pi \hbar^4}}$$

$$= \frac{m\alpha^2}{\pi \hbar^2} - \frac{\alpha \cdot 2m\alpha}{\pi \hbar^2}$$

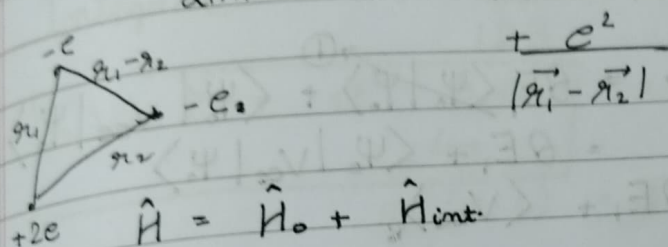
$$= -\frac{m\alpha^2}{\pi \hbar^2}$$

$$E_{gs} \approx -\frac{m\alpha^2}{\pi\hbar^2}$$

$$\text{Exact } E_{gs} = -\frac{m\alpha^2}{2\hbar^2}$$

\* The ground state of Helium atom →

$$\hat{H} = \frac{-\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{r_1} - \frac{2e^2}{r_2}$$



$$\hat{H} = \hat{H}_0 + \hat{H}_{int.}$$

No exact soln. of  $e^- - e^-$  interaction term.

Ignore the  $e^- - e^-$  repulsion term.

G.S. : product of 2 Hydrogenic gs wave f<sup>n</sup>. with  $Z=2$

$$\Psi_{100} \Psi_0(\vec{r}_1, \vec{r}_2) = \Psi_{100}(\vec{r}_1) \cdot \Psi_{100}(\vec{r}_2)$$

$$\Psi_{100}(\vec{r}_1) = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-\frac{Zr_1}{a_0}}$$

$$\Psi_0(\vec{r}_1, \vec{r}_2) = \frac{Z^3}{\pi a_0^3} e^{-\frac{Z(r_1+r_2)}{a_0}}$$

$$E_0 = -13.6 Z^2 \text{ eV for } 1e^-$$

$$\text{Total g.s. energy} = 2(-13.6) Z^2 = 2 \times (-13.6) = -27.2 \text{ eV}$$

$$\text{Actual} = -78.975 \text{ eV}$$

Use trial  $\psi_0: \psi_0 = \frac{B}{\pi a_0^3} e^{-2(r_1 + r_2)/a_0}$

$$\hat{H} = \hat{H}_0 + \hat{H}_{int.}$$

$$\hat{H}_0 = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{r_1} - \frac{2e^2}{r_2}$$

$$\hat{H}_0 \psi_0 = \theta E_1 \psi_0; E_1 = -13.6 \text{ eV}$$

$$\hat{H} \psi_0 = (\hat{H}_0 + \hat{V}_{ee}) \psi_0$$

$$= (\theta E_1 + \hat{V}_{ee}) \psi_0$$

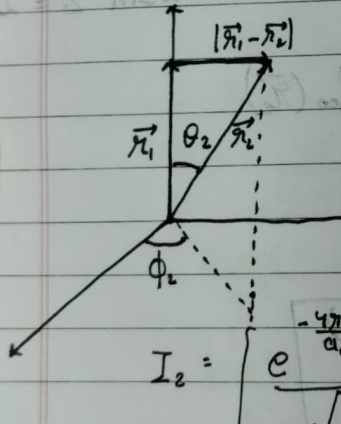
$$\langle \psi_0 | \hat{H} | \psi_0 \rangle = \theta E_1 \langle \psi_0 | \psi_0 \rangle + \langle \psi_0 | \hat{V}_{ee} | \psi_0 \rangle$$

$$= \theta E_1 + \langle \psi_0 | \hat{V}_{ee} | \psi_0 \rangle$$

$$\langle \hat{H} \rangle_{\psi_0} = \theta E_1 + \langle \hat{V}_{ee} \rangle_{\psi_0}$$

$$\langle \hat{V}_{ee} \rangle = \langle \psi_0 | \hat{V}_{ee} | \psi_0 \rangle$$

$$= e^2 \left( \frac{B}{\pi a_0^3} \right)^2 \int \frac{e^{-4(r_1 + r_2)/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3 \vec{r}_1 d^3 \vec{r}_2$$



$$|\vec{r}_1 - \vec{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}$$

$$I_2 = \int \frac{e^{-4r_2/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3 \vec{r}_2$$

$$I_2 = \int \frac{e^{-\frac{4r_2}{a_0}} \cdot r_2^2 \sin \theta_2 dr_2 d\theta_2 d\phi_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}}$$

Integral over  $\theta_2$ :

$$\int_0^\pi \frac{\sin \theta_2 d\theta_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}}$$

$$= \frac{1}{r_1 r_2} \left[ (r_1 + r_2) - |r_1 - r_2| \right]$$

$$= \frac{1}{r_1 r_2} \left[ r_1 + r_2 - r_1 + r_2 \right] \quad r_2 < r_1$$

$$= \frac{1}{r_1 r_2} \left[ 2r_1 \right] \quad r_2 > r_1$$

$$I_2 = 4\pi \left[ \int_0^{r_1} e^{-\frac{4r_2}{a_0}} r_2^2 \cdot \frac{1}{r_2} dr_2 + \int_{r_1}^{\infty} e^{-\frac{4r_2}{a_0}} r_2^2 \cdot \frac{1}{r_2} dr_2 \right]$$

$$I_2 = \frac{\pi a_0^2}{8\lambda_1} \left[ 1 - \left(1 + \frac{2r_1}{a_0}\right) e^{-4r_1/a_0} \right] e^{-4r_1/a_0}$$

$$\langle V_{ee} \rangle = e^2 \left( \frac{\beta}{\pi a_0^3} \right) \int \left[ 1 - \left(1 + \frac{2r_1}{a_0}\right) e^{-\frac{4r_1}{a_0}} \right] e^{-\frac{4r_1}{a_0}} r_1 \sin\theta, dr_1, d\theta, d\phi,$$

$$= e^2 \beta$$

Integral over  $r_1$  :  $\frac{5a_0^2}{128}$

$$\langle V_{ee} \rangle = \frac{5e^2}{4a_0} = \frac{5}{2} (13.6) = 34 \text{ eV}$$

$$\langle H \rangle = -109 + 34 = -75 \text{ eV}$$

Take  $Z$  as a parameter in trial f.m.

$$\psi_0 = \frac{Z^3}{\pi a_0^3} e^{-\frac{Z(r_1+r_2)}{a_0}} \hat{u}_0$$

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2)}_{\text{unchanged}} - \underbrace{e^2 \left( \frac{Z}{r_1} + \frac{Z}{r_2} \right)}_{\hat{H}_1} + \underbrace{\frac{e^2}{|\vec{r}_1 - \vec{r}_2|}}_{\hat{V}_{e-e}}$$

$$\begin{aligned}
 \langle H \rangle &= \langle H_0 \rangle + \langle H_1 \rangle + \langle V_{ee} \rangle \\
 &= 2E_1 Z^2 + 2(Z-2)Z \frac{e^2}{a_0} + \frac{5Z e^2}{4} \\
 &= 2E_1 Z^2 + 4(Z-2)Z (-E_1) + \frac{5Z}{4} (-E_1) \\
 &= E_1 \left( -2Z^2 + \frac{27}{4} Z^2 \right)
 \end{aligned}$$

Minimize  $\langle H \rangle$  w.r.t  $Z \Rightarrow \frac{d\langle H \rangle}{dZ} = 0$

$\Rightarrow Z = 1.69$

Plugging in  $\langle H \rangle \rightarrow$

$\langle H \rangle = -77.5 \text{ eV}$

\* System of 2-spin- $\frac{1}{2}$  particles

Particle A:  $|\uparrow\rangle_A, |\downarrow\rangle_B$

Particle B:  $|\uparrow\rangle_B, |\downarrow\rangle_B$

States of combined system of particle A & B

①

$|\uparrow\rangle_A \otimes |\uparrow\rangle_B \equiv |\uparrow\uparrow\rangle$

↳ Tensor Product

②

$(\alpha_A |\uparrow\rangle_A + \beta_A |\downarrow\rangle_A) \otimes (\alpha_B |\uparrow\rangle_B + \beta_B |\downarrow\rangle_B)$

$= \alpha_A \alpha_B |\uparrow\rangle_A \otimes |\uparrow\rangle_B + \alpha_A \beta_B |\uparrow\rangle_A \otimes |\downarrow\rangle_B + \beta_A \alpha_B |\downarrow\rangle_A \otimes |\uparrow\rangle_B + \beta_A \beta_B |\downarrow\rangle_A \otimes |\downarrow\rangle_B$

③

$|\uparrow\rangle_A \otimes |\downarrow\rangle_B + |\downarrow\rangle_A \otimes |\uparrow\rangle_B$

$\Rightarrow \alpha_A \alpha_B = 0$

$\beta_A \beta_B = 0$

$\alpha_A \beta_B = 1$

$\beta_A \alpha_B = 1$

$\alpha_A \neq 0$

$\beta_A \neq 0$

$\beta_B \neq 0$

$\alpha_B \neq 0$

Entangled State.

NOT POSSIBLE  $\Rightarrow$  NOT a product state